CO4 - AT2 - INDUSTRY PROBLEM SOLVING

Course Code: ITA0609

Course Title: Machine Learning

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1. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, howrecommendations are generated, and how the system adapts to changing user behavior over time.

Sol:-**Introduction**

A telecommunications company can use the **K-Nearest Neighbour (KNN)** algorithm to recommend the most suitable data plan to customers based on their usage patterns. The main idea is to find customers who have **similar usage behaviour** to a new customer and recommend the data plan that is commonly used by those similar customers.

The system can consider factors such as monthly data usage, number of calls, SMS usage, video streaming, social media usage, and monthly spending. Since KNN is a simple and instance-based learning algorithm, it can easily adapt when new customer data becomes available.

1. Data Collection

First, the company collects historical customer information. The dataset may contain the following features:

Customer	**Data Usage (GB)**	**Calls/Month**	**SMS/Month**	**Streaming (GB)**	**Monthly Bill (₹)**	**Plan**
C1	2	100	80	0.5	199	Basic
C2	4	180	120	1.0	299	Basic
C3	6	220	150	2.0	399	Standard
C4	10	300	200	4.0	599	Premium
C5	12	350	250	5.0	699	Premium

When a new customer joins the network, the system collects the same type of information.

For example, a new customer may have:

- Data usage = 9 GB/month
- Calls = 290/month
- SMS = 190/month

- Video streaming = 3.5 GB/month

2. Measuring Similarity Between Customers

KNN determines how similar the new customer is to existing customers by calculating the **distance** between their feature values.

The most commonly used method is **Euclidean distance**.

The formula is:

Distance $= \sqrt{[(x_1-y_1)^2 + (x_2-y_2)^2 + \ldots + (x_n-y_n)^2]}$

Here:

- x represents the usage values of the new customer.
- y represents the usage values of an existing customer.
- A **smaller distance** means that the customers have more similar usage patterns.
- A **larger distance** means that their usage patterns are different.

Since features such as data usage, calls, and SMS have different ranges, the values should first be **normalized or standardized**. This prevents one feature from having an unnecessarily large influence on the distance calculation.

3. Selecting the Nearest Customers

After calculating the distances, the system sorts all existing customers according to their distance from the new customer.

Suppose the company selects **K = 3**. The system chooses the three customers with the smallest distances.

For example:

Nearest Customer	Distance	Existing Plan
C4	0.35	Premium
C5	0.48	Premium
C3	1.20	Standard

The three nearest customers use:

- Premium Plan → 2 customers
- Standard Plan → 1 customer

Therefore, the system recommends the **Premium Plan** to the new customer.

4. Generating the Recommendation

The recommendation is generated using **majority voting** among the K nearest neighbours.

For example, if K = 5 and the plans of the nearest customers are:

Premium, Premium, Standard, Premium, Standard

Then:

- Premium = 3 customers
- Standard = 2 customers

Therefore, **Premium** is selected as the recommended plan.

The complete process can be represented as:

Customer Usage Data → Data Preprocessing → Feature Normalization → Calculate Distance → Find K Nearest Customers → Majority Voting → Recommend Data Plan

5. Example Scenario

Consider a customer whose current usage is approximately **9 GB per month**, with high video streaming and around **290 calls per month**.

The system compares this customer with thousands of existing customers. It finds that most customers with similar usage patterns have selected high-data plans.

If the nearest customers mostly use a **Premium Plan**, the system recommends the Premium Plan.

The company can also display additional information such as:

- Recommended plan
- Monthly data allowance
- Expected monthly cost
- Benefits of the plan
- Comparison with the customer's current plan

This makes the recommendation more useful to the customer.

6. Adapting to Changing Customer Behaviour

One important advantage of a KNN-based system is that it can adapt to changing customer behaviour.

Customer usage is not always constant. For example, a customer may initially use only **2 GB/month**, but after starting to watch more online videos, their usage may increase to **8–10 GB/month**.

The system continuously collects updated usage information.

For example:

Initial usage: 2 GB/month → Basic Plan

After 3 months: 5 GB/month → Standard Plan

After 6 months: 10 GB/month → Premium Plan

When new usage data is added to the database, KNN can use this updated information when making future recommendations. The company can also give greater importance to **recent usage patterns**, so old behaviour does not strongly affect the current recommendation.

7. Handling New Customer Behaviour

The system can periodically update the customer database with:

- Recent data consumption
- Recent call usage
- Recent SMS usage
- Streaming behaviour
- Changes in monthly spending
- Previously accepted or rejected recommendations

8. Advantages of the KNN Approach

1. **Simple to implement:** KNN is easy to understand and implement.
2. **Personalized recommendations:** Plans are recommended according to individual usage.
3. **No complex training process:** KNN stores existing customer data and uses it during prediction.
4. **Adapts to new data:** New customer records can be added easily.
5. **Suitable for changing behaviour:** Updated usage patterns can influence future recommendations.
6. **Effective for customer segmentation:** Customers with similar behaviour can be grouped together.

9. Limitations

KNN can become slower when the company has a very large number of customers because it needs to calculate distances between the new customer and many existing customers. The system also requires proper data preprocessing and selection of an appropriate value of **K**.

Conclusion

A **K-Nearest Neighbour based data-plan recommendation system** can help a telecommunications company provide personalized plans to its customers. The system collects customer usage information and measures similarity using **Euclidean distance** after normalization. It then identifies the **K most similar customers** and recommends the plan that receives the highest number of votes. By continuously updating the customer database with recent usage behaviour, the system can adapt to changes in customer requirements. Thus, KNN provides a simple, flexible, and effective approach for personalized telecommunications data-plan recommendations.

2. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

Sol:-**Introduction**

An energy management company can use **Locally Weighted Regression (LWR)** to predict electricity consumption based on factors such as **time of day, weather conditions, temperature, and appliance usage**. LWR is suitable for this problem because electricity consumption does not always follow one fixed relationship. For example, electricity usage during the morning may be different from usage during the afternoon or night.

Unlike ordinary linear regression, which tries to learn one global relationship for the entire dataset, LWR gives **more importance to data points that are close to the current prediction point**. Therefore, it can capture local variations in electricity consumption patterns.

1. Data Collection

The first step is to collect historical electricity consumption data. The dataset can contain:

- Time of day
- Day of the week
- Month or season
- Temperature
- Humidity
- Weather condition
- Number of appliances being used
- Air-conditioner usage
- Refrigerator usage
- Previous electricity consumption
- Total electricity consumption

For example:

Time	Temperature (°C)	AC Usage (hrs)	Appliances	Previous Usage (kWh)	Consumption (kWh)
06:00	24	0.5	4	2.1	2.3
09:00	27	1.0	6	2.5	2.8
13:00	33	3.0	8	4.0	5.2
18:00	30	2.0	9	4.5	5.0

22:00	27	1.0	5	3.2	3.4

The consumption value is the **target variable**, while the remaining factors are used as input features.

2. Preprocessing the Data

Before applying LWR, the collected data is cleaned and prepared.

The company can:

- Remove duplicate records.
- Handle missing values.
- Remove or reduce the effect of extreme outliers.
- Normalize numerical features.
- Convert time and seasonal information into suitable numerical features.

For example, the hour of the day can be represented using **sine and cosine transformations** so that 23:00 and 00:00 are treated as nearby times rather than completely different values.

3. Local Learning in LWR

The main idea of LWR is **local learning**.

Suppose the company wants to predict electricity consumption at **2:00 PM**. Instead of giving equal importance to every historical record, LWR gives higher weights to records that are similar to 2:00 PM.

For example:

- Historical data around 1:00–3:00 PM → **high weight**
- Data around 11:00 AM–1:00 PM → **moderate weight**
- Data from midnight → **low weight**

The weight can be calculated using a Gaussian kernel:

$$\mathbf{w_i = \exp(-d_i^2 / 2\tau^2)}$$

where:

- w_i = weight of the data point
- d_i = distance between the current input and historical data point
- τ = bandwidth parameter
- A smaller distance gives a higher weight.

Thus, nearby and more relevant observations have a greater influence on the prediction.

4. How Local Learning Captures Consumption Variations

Electricity consumption changes throughout the day and across different conditions.

For example:

Morning:
People wake up and use lights, water heaters, kitchen appliances, etc., causing consumption to increase.

Afternoon:
Temperature may increase, leading to higher air-conditioner usage.

Evening:
People return home, increasing the use of lights, televisions, fans, computers, and other appliances.

Night:
Most appliances are switched off, so consumption generally decreases.

A global regression model may struggle to represent all these different patterns with a single equation. LWR instead learns a separate **local approximation around each prediction point**, allowing it to capture these variations more effectively.

5. Handling Weather Effects

Weather has a significant influence on electricity consumption.

For example, when the temperature increases:

Higher temperature → More AC/Fan usage → Higher electricity consumption

Similarly, during cooler weather, air-conditioner usage may decrease.

LWR can identify historical observations with similar temperature and weather conditions and give those observations higher weights. Therefore, the prediction can reflect the effect of current weather conditions.

6. Handling Seasonal Data

Electricity consumption can also vary according to seasons.

For example:

- **Summer:** Higher AC and fan usage.
- **Winter:** Lower cooling requirements but possibly higher heating usage.
- **Monsoon:** Different lighting and cooling patterns.
- **Weekends:** Different household usage compared with weekdays.

The model can include **month, season, day of the week, and time of day** as input features. When predicting consumption during summer, LWR gives greater importance to historical observations from similar summer conditions.

This allows the model to capture **seasonal patterns** without requiring one single global relationship for the entire year.

7. Handling Noisy Data

Real-world electricity data may contain noise because of:

- Sudden appliance switching
- Power failures
- Sensor errors
- Unexpected weather changes
- Unusual customer behaviour
- Faulty measurements

LWR can reduce the effect of irrelevant observations because distant data points receive smaller weights. The company can also improve robustness by:

- Removing extreme outliers.
- Using smoothing techniques.
- Choosing an appropriate bandwidth.
- Using regularization when fitting the local regression model.

The **bandwidth parameter (τ)** is particularly important. If the bandwidth is too small, the model may follow noise too closely. If it is too large, the model becomes too similar to global regression and may miss local patterns.

8. Prediction Process

The complete prediction process is:

Historical Energy Data
↓
Data Cleaning and Normalization
↓
Select Relevant Features
↓
Calculate Distance from Current Input
↓
Assign Local Weights
↓
Perform Weighted Regression
↓
Predict Electricity Consumption

For example, if the current conditions are:

- Time = 2:00 PM
- Temperature = 34°C
- AC usage = 3 hours
- Appliances = 8

LWR searches for historical observations with similar time, temperature, and appliance usage. These observations receive higher weights, and a local regression model is used to estimate the expected electricity consumption.

9. Advantages of LWR

1. **Captures local patterns:** It can learn different relationships for different times and conditions.
2. **Handles nonlinear behaviour:** Electricity consumption does not have to follow one straight-line relationship.
3. **Adapts to changing conditions:** New observations can influence future predictions.
4. **Useful for seasonal data:** Time and seasonal factors can be included as features.
5. **Reduces the effect of distant observations:** Local weights prevent unrelated historical data from strongly affecting predictions.
6. **Suitable for energy management:** It can help companies forecast demand and manage electricity resources efficiently.

10. Applications

The predicted electricity consumption can be used for:

- Electricity demand forecasting
- Smart-grid management
- Peak-load prediction
- Energy-efficient appliance scheduling
- Dynamic energy management
- Reducing electricity wastage
- Planning power generation and distribution

Conclusion

Locally Weighted Regression (LWR) provides an effective solution for predicting electricity consumption using factors such as time, weather, temperature, and appliance usage. Its main advantage is **local learning**, where nearby and similar historical observations receive higher weights than unrelated observations. This enables the model to capture different consumption patterns during mornings, afternoons, evenings, weekends, and different seasons. Noise can be controlled through preprocessing, outlier handling, and suitable selection of the bandwidth parameter. Therefore, LWR is a flexible approach for accurate and adaptive electricity consumption prediction in an energy management system.